Technical Report on AI and Multimedia Authenticity Standards

Mapping the standardisation landscape

2026-06-15

Executive Summary

This technical report provides a comprehensive overview of the current landscape of standards and specifications related to digital media authenticity and artificial intelligence. It categorizes these standards into five key clusters: content provenance, trust and authenticity, asset identifiers, rights declarations, and watermarking. The report provides a short description of each standard along with links for further details.

By mapping the contributions of various Standard Development Organizations (SDOs) and groups, we aim to identify gaps and opportunities for further standardization. This could serve as a valuable resource for stakeholders seeking to navigate the complex ecosystem of standards at the intersection of artificial intelligence and authenticity in digital media and to implement best practices to safeguard the authenticity of digital assets and the rights assigned to them. The findings underscore the critical role of robust specifications and standards in fostering trust and accountability in the evolving digital landscape.

AI and Multimedia Authenticity Standards Collaboration (AMAS)

The AI and Multimedia Authenticity Standards Collaboration is a global initiative advancing standardization in the rapidly evolving field of AI-generated and altered media. By identifying gaps and driving the development of new standards, we support transparent, privacy-conscious, and rights-respecting practices. Our work also aims to inform policy and regulatory frameworks to promote legal compliance and safeguard public trust.

Led by the World Standards Cooperation, the collaboration serves as a vital forum for dialogue among standards developers, civil society organizations, technology companies, and other key players. Participating organizations include the International Electrotechnical Commission (IEC), the International Organization for Standardization (ISO), the International Telecommunication Union (ITU), the Coalition for Content Provenance and Authenticity (C2PA), the China Academy of Information and Communications Technology (CAICT), DataTrails, Deep Media, and Witness.

Convened by ITU under the auspices of the World Standards Cooperation, the collaboration was launched at the AI for Good Global Summit in 2024.

Learn more here (https://aiforgood.itu.int/multimedia-authenticity/) or contact the Secretariat at amas-secretariat@itu.int

Disclaimer

This report is a collaborative work prepared by the secretariats of the International Electrotechnical Commission (IEC), the International Organization for Standardization (ISO), and the International Telecommunication Union (ITU) under the banner of the World Standards Cooperation (WSC).

The views, observations, and conclusions expressed in this publication are solely those of the authors, including from the respective secretariats. They do not necessarily reflect, nor do they represent, the official positions, policies, or consensus views of the national member bodies, or any other affiliated members of IEC, ISO, or ITU.

This document is intended to provide a technical overview and mapping of the standardization landscape concerning AI and multimedia authenticity for informational purposes. It has not been subject to the formal approval processes of these standards development organizations and should not be construed as an official standard or a formal endorsement by their respective membership.

Introduction

This technical paper provides a comprehensive overview of various standards and specifications related to digital media, focusing on content provenance, trust and authenticity, asset identifiers, rights declarations, and watermarking. These standards are essential for ensuring the authenticity of both synthetic and non-synthetic digital content. As generative AI continues to evolve, the need for robust standards becomes increasingly critical to protect the interests of creators, consumers, and organizations.

This technical report also contains a gap analysis categorized into governance, access, technical performance, and societal impact.

The standards discussed in this report are developed by various Standard Development Organizations (SDOs) and groups, each contributing to different aspects of AI and authenticity. By adhering to these guidelines, organizations can better maintain the trustworthiness and provenance of their digital assets, ensuring that content remains authentic and traceable throughout its lifecycle.

Methodology

The approach taken in this report involves a thorough review of existing standards and specifications related to digital media. The focus is on identifying key areas where standards already exist, and from that, what is still needed to ensure the authenticity and integrity of digital content. From that review, we have broken this report into the categories: content provenance, trust and authenticity, asset identifiers, rights declarations, and watermarking.

In addition to the above, sessions were organized during AI for Good Summit 2025 and AI Standards Summit 2025 during which stakeholders were consulted. This resulted in a gap analysis of the landscape of the current ongoing standardization efforts and their challenges.

Categories of Standards

In this technical paper, we have clustered those standards and specifications in the scope of this analysis into five categories: content provenance, trust and authenticity, asset identifiers, rights declaration and watermarking. Rights declaration, in turn, is defined in two inclinations: general purpose and opt-out mechanisms. The general-purpose rights declaration addresses a broad scope while the opt-out mechanisms refer to a specific aspect of rights declaration that is of relevance to the scope of this document.

Content Provenance

Content provenance refers to information on the origin, history and lifecycle of digital content. This is an important tool for the verification of the authenticity and integrity of digital assets. Provenance information helps in tracking the creation, modification, and distribution of content, providing a transparent record that can be used to establish trust and accountability. This is particularly important in use cases and applications where the authenticity of content is paramount, such as in journalism, scientific research, and digital art.

Trust and Authenticity

Trust and authenticity measures ensure that digital content is genuine and has not been tampered with. Such mechanisms are essential for maintaining the integrity of digital media, especially in environments where content manipulation is a significant concern. By implementing trust and authenticity measures, organizations can protect their digital assets from unauthorized alterations and ensure that consumers can rely on the content they receive.

Asset Identifiers

Asset identifiers are unique codes assigned to digital content to ensure proper management and identification of asset. These identifiers help in maintaining a clear and organized record of digital assets, making it easier to organize, manage and distribute content. By using asset identifiers, organizations can ensure that their digital media is properly tracked and accounted for, reducing the risk of loss or unauthorized use.

Rights Declarations

General Purpose

Rights declarations are formal statements that outline the rights and permissions associated with digital content. These declarations help in clarifying the ownership and usage rights of digital assets, providing a clear framework for how content can be used and shared. By establishing clear rights declarations, organizations can protect their intellectual property and ensure that their digital assets are used in accordance with their intended purpose.

Opt-Out Mechanisms

Opt-out mechanisms are a specialized approach to rights declarations that allow users to exclude their content from certain processes, such as data mining or AI training. These mechanisms are essential for protecting the privacy and rights of content creators, ensuring that their digital assets are not used without their consent. By implementing opt-out mechanisms, organizations can provide users with greater control over their content and ensure that their rights are respected.

Watermarking

Watermarking ensures that digital content is marked in a way that can be used to verify its authenticity, synthetic (or not) nature, and ownership. Watermarking is increasingly used to facilitate the declaration of the rights of content creators and ensuring that their digital assets are not used without their consent. By implementing watermarking measures, organizations can provide users with greater control over their content and make sure that their rights are respected.

Overview of Specifications

Content Credentials (C2PA)

- **SDO/Group:** C2PA
- **Link:** C2PA Specification
- **Status:** Published
- **Publication Date:** 2026
- **Media:** Any
- **Summary:** This standard focuses on providing a tamper-evident provenance record, called a Content Credential, that can be embedded into digital media, watermarked or stored online. A Content Credential can include information about the creator, creation date, and any modifications made to the content. This helps in maintaining a verifiable record of the content's history.

JPEG Trust Part 1: Core foundation

- **SDO/Group:** ISO/IEC JTC 1/SC 29/WG 1
- **Link:** ISO 21617-1:2025, second edition in progress
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Any (image focused)
- **Summary:** This standard focuses on trust and authenticity in JPEG images through provenance, detection and fact-checking. It provides a framework for embedding metadata directly into JPEG files in the form of trust indicators. This allows users to decide the degree of trust they can put on a digital asset as a function of their trust profiles. This is particularly useful in contexts where image manipulation is common, such as in social media applications.

JPEG Trust Part 2: Trust profiles catalog

- **SDO/Group:** ISO/IEC JTC 1/SC 29/WG 1
- **Link:** ISO 21617-2
- **Status:** In Progress
- **Media:** Any (image focused)
- **Summary:** This standard introduces trust indicator sets, trust profiles and trust reports that can be used to in specific workflows, use cases and applications such as broadcasting, digital cameras, AI-powered content generation services, etc.

JPEG Trust Part 3: Media asset watermarking

- **SDO/Group:** ISO/IEC JTC 1/SC 29/WG 1
- **Link:** ISO 21617-3
- **Status:** In Progress
- **Media:** Images
- **Summary:** This standard provides an overview of mechanisms used for watermarking of media assets.

CAWG Metadata

- **SDO/Group:** Creation Assertions Working Group, as part of DIF
- **Link:** CAWG Metadata
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Any
- **Summary:** This specification provides a framework for expressing metadata that captures detailed information about the content, including ownership and authorship.

Originator Profile

- **SDO/Group:** Originator Profile
- **Link:** Originator Profile
- **Status:** In progress

- **Media:** Web pages
- **Summary:** This specification provides a framework for documenting the origin of digital content. It includes guidelines for creating and maintaining profiles that capture detailed information about the content's creator and its creation process. This helps in establishing a clear and verifiable record of the content's provenance.

Overview of trustworthiness in artificial intelligence

- **SDO/Group:** ISO/IEC JTC 1/SC 42
- **Link:** ISO/IEC TR 24028:2020
- **Status:** Published
- **Publication Date:** 2020
- **Summary:** This standard offers an overview of trustworthiness in artificial intelligence. It provides guidelines for assessing the reliability and integrity of AI systems, ensuring that they produce trustworthy results. This is crucial in applications where AI is used to generate or manipulate digital content, as it helps in maintaining the authenticity of the output.

Framework for trust-based media services

- **SDO/Group:** ITU-T
- **Link:** ITU-T Y.3054
- **Status:** Published
- **Publication Date:** 2018
- **Summary:** is framework provides guidelines for trust-based media services. In particular, it includes methods for establishing and maintaining trust in digital media platforms, ensuring that users can rely on the content they access. This is particularly important in contexts where media services are used to distribute sensitive or high-value content.

Trust.txt

- **SDO/Group:** JournalList
- **Link:** Trust.txt
- **Status:** Initiated
- **Media:** Web pages

- **Summary:** This specification outlines methods for establishing trust in digital content. It includes guidelines for creating and maintaining trust.txt files, which can be used to document the trustworthiness of digital assets. This helps in ensuring that users can verify the authenticity of the content they receive.

Chromium Reputation Provider Framework

- **SDO/Group:** Google's Chrome Team
- **Link:** Chromium Reputation Provider Framework
- **Status:** Initiated
- **Media:** Web pages
- **Summary:** This framework provides guidelines for reputation management of digital content. It includes methods for assessing and maintaining the reputation of digital assets, ensuring that users can trust the content they access. This is particularly important in contexts where reputation is a key factor in determining the value and reliability of digital media.

ISCC: International Standard Content Code (ISCC)

- **SDO/Group:** ISO/TC 46/SC 9
- **Link:** ISO 24138:2024
- **Status:** Published
- **Publication Date:** 2024
- **Media:** Any
- **Summary:** This standard provides a unique identifier for digital content. It includes guidelines for creating and maintaining ISCC codes, which can be used to track and manage digital assets. This helps in ensuring that content is properly accounted for and can be easily identified and retrieved.

Global Media Identifier (GMI)

- **SDO/Group:** IWA 44
- **Link:** GMI
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Any

- **Summary:** This specification offers a unique identifier for media content. It includes methods for creating and maintaining UMid codes, which can be used to track and manage media assets. This helps in ensuring that content is properly accounted for and can be easily identified and retrieved.

TDM Reservation Protocol

- **SDO/Group:** W3C
- **Link:** TDMRep
- **Status:** Published
- **Publication Date:** 2024
- **Media:** Web pages, EPUB, PDF
- **Summary:** This protocol provides guidelines for reserving content from text and data mining. It includes methods for creating and maintaining TDMRep files, which can be used to document the reservation of digital assets. This helps in ensuring that content is not used for data mining without the creator's consent.

Robots Exclusion Protocol

- **SDO/Group:** IETF
- **Link:** RFC 9309
- **Status:** Published
- **Publication Date:** 2022
- **Media:** Any
- **Summary:** This standard provides guidelines for excluding content from web crawlers. It includes methods for creating and maintaining robots.txt files, which can be used to document the exclusion of digital assets. This helps in ensuring that content is not accessed by web crawlers without the creator's consent.

Vocabulary for expressing AI usage preferences

- **SDO/Group:** IETF
- **Link:** draft-ietf-aipref-vocab-06
- **Status:** In progress
- **Media:** Any

- **Summary:** This document proposes a standardized vocabulary of use cases that can be targeted when expressing machine-readable opt-outs related to Text and Data Mining (TDM) and AI training. The vocabulary is agnostic to specific opt-out mechanisms and enables declaring parties to communicate restrictions or permissions regarding the use of their digital assets in a structured and interoperable manner.

Open Binding of Content Identifiers (OBID)

- **SDO/Group:** SMPTE
- **Link:** SMPTE ST 2112-10:2020
- **Status:** Published
- **Publication Date:** 2020
- **Media:** Audio
- **Summary:** This standard provides guidelines for binding content identifiers to digital media. It includes methods for creating and maintaining OBID files, which can be used to document the binding of digital assets. This helps in ensuring that content is properly accounted for and can be easily identified and retrieved.

X.ig-dw: Implementation guidelines for digital watermarking

- **SDO/Group:** ITU-T SG17
- **Link:** SG17-TD42/WP4
- **Status:** Published, but temporary
- **Publication Date:** 2024
- **Media:** Images, Video
- **Summary:** This guideline offers methods for implementing digital watermarking. It includes guidelines for creating and maintaining watermark files, which can be used to document the watermarking of digital assets. This helps in ensuring that content is properly accounted for and can be easily identified and retrieved.

DRM technology for digital publications Part 1: Overview of copyright protection technologies in use in the publishing industry

- **SDO/Group:** ISO/IEC JTC 1/SC 34
- **Link:** ISO/IEC 23078-1:2024

- **Status:** Published
- **Publication Date:** 2024
- **Media:** EPUB, PDF
- **Summary:** This standard provides an overview of DRM technologies for digital publications. It includes guidelines for creating and maintaining DRM files, which can be used to document the DRM of digital assets. This helps in ensuring that content is properly accounted for and can be easily identified and retrieved.

IEEE Draft Standard for Evaluation Method of Robustness of Digital Watermarking Implementation in Digital Contents

- **SDO/Group:** IEEE
- **Link:** IEEE P3361
- **Status:** In progress
- **Media:** Any
- **Summary:** This draft standard offers methods for evaluating the robustness of digital watermarking. It includes guidelines for creating and maintaining evaluation files, which can be used to document the evaluation of digital assets. This helps in ensuring that content is properly accounted for and can be easily identified and retrieved.

H.MMAUTH: Framework for authentication of multimedia content

- **SDO/Group:** ITU-T SG21/Q9
- **Link:** H.MMAUTH
- **Status:** In progress
- **Media:** Video
- **Summary:** This Draft Recommendation specifies a technical solution for the verification of multimedia content integrity, enabling users to confirm the authenticity of the content by its creators, such as governments, companies, or news organizations. The solution is based on the digital signing of data streams. The content creator (encoder) uses a private key to sign the content, while the recipient (decoder) uses a corresponding public key to verify the authenticity. The public key, necessary for verification, is not derived directly from the data stream but is obtained through a trusted, independent method, such as a third-party trust center.

H.274(V4): Versatile supplemental enhancement information messages for coded video bitstreams

- **SDO/Group:** JVET (ITU-T SG21 & ISO/IEC JTC 1/SC 29/ WG5)
- **Link:** H.274(V4)
- **Status:** Published
- **Publication Date**: 2026
- **Media:** Video
- **Summary:** This specification contains changes to the versatile supplemental enhancement information messages for coded video bitstreams (VSEI) standard to specify additional SEI messages that will be useful for the purposes of content provenance, trust and authenticity.

H.VADS: Assessment criteria for video authenticity detection services

- **SDO/Group:** ITU-T SG21/Q7
- **Link:** H.VADS
- **Status:** In progress
- **Media:** Video
- **Summary:** This Draft Recommendation provides a comprehensive assessment framework for video authenticity detection services. It specifies the requirements, assessment categories, key metrics, and methods to evaluate the capabilities of video authenticity detection services. By establishing a structured, criteria-based approach, this Draft Recommendation would guide the development, evaluation, and selection of reliable and effective video authenticity detection services.

Credible Web

- **SDO/Group:** W3C
- **Link:** Cred Web
- **Status:** In Progress
- **Media:** Web pages
- **Summary:** The mission of the W3C Credible Web Community Group is to help shift the Web toward more trustworthy content without increasing censorship or social division. We want users to be able to tell when content is reliable, accurate, and

shared in good faith, and to help them steer away from deceptive content. At the same time, we affirm the need for users to find the content they want and to interact freely in the communities they choose. To balance any conflict between these goals, we are committed to providing technologies which keep end-users in control of their Web experience.

Technical and Governance Guidelines for Responsible Data Collection

- **SDO/Group:** Alliance for Responsible Data Collection (ARDC)
- **Link:** Technical and Governance Guidelines for Responsible Data Collection
- **Status:** Published
- **Publication Date:** 2024
- **Media:** Data
- **Summary:** With the rapid expansion of online digital data, it is critical to establish responsible data collection standards that provide data collectors with guidance on best practices, provide third parties with a reliable means to assess whether the public web data they seek to use has been responsibly sourced, and protect public access to public data. The technical guidelines set forth below are part of a broader framework for responsible data collection that includes important guidelines for data collection governance. As such, all ARDC guidelines must be applied holistically. These standards build upon previous work by the FISD-SIA-Alternative Data Council on Web Data Collection Considerations.

Data Provenance Standards

- **SDO/Group:** Data & Trust Alliance
- **Link:** D&TA Data Provenance Standards v1.0.0
- **Status:** Published
- **Publication Date:** 2024
- **Media:** Data
- **Summary:** The Data Provenance Standards are a set of guidelines developed by the Data & Trust Alliance to help organizations assess dataset quality, transparency, and legal usability for AI and traditional data applications. The standards aim to provide essential metadata about a dataset's origin, creation method, and legal usage, with the goal of increasing data trust and reducing risks in AI development.

IEEE Standard for Transparent Human and Machine Agency Identification

- **SDO/Group:** IEEE
- **Link:** IEEE 3152-2024
- **Status:** Published
- **Publication Date:** 2024
- **Media:** Images, Audio
- **Summary:** This standard addresses recognizable audio and visual markers that assist humans in distinguishing communication with a human, a machine, or a combination of both. Therefore, the standard defines visual, textual, and auditory marks. This standard does not cover methods to determine whether an interaction is with a machine, such as Turing tests.

IPTC Photo Metadata Standard

- **SDO/Group:** IPTC
- **Link:** IPTC Photo Metadata Standard 2025.1
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Images, Video
- **Summary:** This document specifies metadata properties with a focus on usage with photos, some of these properties are also specified by the IPTC Video Metadata Hub.

DASH - Part 4: Segment Encryption and Authentication

- **SDO/Group:** ISO/IEC JTC 1/SC 29
- **Link:** ISO/IEC 23009-4:2018
- **Status:** Published
- **Publication Date:** 2018
- **Media:** Video
- **Summary:** This standard introduces encryption and authentication mechanisms at the segment level for adaptive video streaming, which enhances content integrity

and protects against tampering during media transmission. It complements existing watermarking and trust/authenticity standards, especially for streaming use cases.

Recommended Practices for Levels of Artificial Intelligence Generated Content Technologies

- **SDO/Group:** IEEE SA
- **Link:** P3429
- **Status:** Published
- **Publication Date:** 2023
- **Media:** Any
- **Summary:** This recommended practice offers a structured framework for understanding and classifying Artificial Intelligence Generated Content (AIGC). It defines rules and levels of AIGC technologies, outlines recommended practices for their implementation, and provides real-world use cases. This standard is highly relevant to the trust and authenticity domain, as it supports transparent communication of the origin, nature, and reliability of AI-generated content—an increasingly critical aspect of digital media governance.

CAWG Identity Assertion Specification

- **SDO/Group:** Creation Assertions Working Group (DIF)
- **Link:** CAWG Identity Assertion v1.2
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Any
- **Summary:** This specification enables content creators to cryptographically bind their verified identity to C2PA-signed digital assets. It supports multiple identity binding methods including X.509 certificates and Identity Claim Aggregators (ICA). Ratified by the Decentralized Identity Foundation (DIF) on December 15, 2025, after CAWG formally became a DIF working group in March 2025. Directly extends and complements the C2PA Specification and CAWG Metadata specification.

W3C Verifiable Credentials Data Model v2.0

- **SDO/Group:** W3C
- **Link:** VC Data Model 2.0

- **Status:** Published
- **Publication Date:** 2025
- **Media:** Any
- **Summary:** This W3C Recommendation (published May 15, 2025) provides a standard data model for cryptographically secure, privacy-respecting, machine-verifiable credentials. It is directly applicable to content authenticity workflows ‚Äî notably as the foundation for identity binding in CAWG Identity Assertions. The v2.0 family comprises seven interlocking specifications covering the data model, JSON-LD contexts, and securing mechanisms (ECDSA and BBS). Supersedes v1.1.

NIST AI 100-4: Reducing Risks Posed by Synthetic Content

- **SDO/Group:** NIST
- **Link:** NIST AI 100-4
- **Status:** Published
- **Publication Date:** 2024
- **Media:** Images, Audio, Video, Any
- **Summary:** This NIST technical report provides authoritative US government guidance on detecting, authenticating, and labeling AI-generated and synthetic content. It covers digital watermarking, metadata-based provenance, and content labeling approaches, with explicit interoperability references to C2PA and W3C PROV as recommended standards. Although a technical report rather than a normative standard, it carries significant influence on US policy and industry practice for AI content authenticity.

MPEG-21 Part 3: Digital Item Identification

- **SDO/Group:** ISO/IEC JTC 1/SC 29
- **Link:** ISO/IEC 21000-3:2025
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Any
- **Summary:** This 2025 edition of MPEG-21 Part 3 replaces the original 2003 version and defines a framework for the unique identification of digital items and their components. It provides persistent, globally unique identifiers for digital content

objects and is applicable across audio, video, images, and documents. Relevant to asset identifier workflows and interoperable with C2PA, ISCC, and GMI (IWA 44) identification schemes.

Cybersecurity technology—Labeling method for content generated by artificial intelligence

- **SDO/Group:** Chinese National Standard
- **Link:** GB 45438
- **Status:** Published
- **Publication Date:** 2025
- **Media:** Images, Audio, Video
- **Summary:** This document defines the compulsory technical methods for explicit and implicit marking of AI-generated text, images, audio, and video.

ITU-T X.2210

- **SDO/Group:** ITU-T SG17
- **Link:** ITU-T X.2210
- **Status:** In Progress

HSTP-MMAuth: Multimedia authentication framework, workflows and procedures

- **SDO/Group:** ITU-T SG21
- **Link:** HSTP-MMAuth
- **Status:** In Progress
- **Media:** Video
- **Summary:** This report will provide an end-to-end framework for multimedia authentication for various use cases such as broadcast, broadband distributions and live/low latency/on-demand streaming, real-time communication, ingesting the content for production and consumption by professional and public consumers. The framework will describe the workflows for various use cases, the role of each actor and entity, the interface requirement between various entities, and the requirements for each entity’s functionalities and features.

Standardization Mapping

Standards & Specification Map

	Content Provenance	Trust and Authenticity	Asset Identifiers	Rights Declarations	Watermarking	Other
Content Credentials (C2PA)	●	●			●	
JPEG Trust Part 1: Core foundation	●	●	●	●	●	
JPEG Trust Part 2: Trust profiles catalog		●				
JPEG Trust Part 3: Media asset watermarking					●	
CAWG Metadata	●	●		●		
Originator Profile	●	●				
Overview of trustworthiness in artificial intelligence		●				
Framework for trust-based media services		●				
Trust.txt		●		●		
Chromium Reputation Provider Framework		●				
ISCC: International Standard Content Code (ISCC)			●			
Global Media Identifier (GMI)			●			
TDM Reservation Protocol				●		
Robots Exclusion Protocol				●		
Vocabulary for expressing AI usage preferences				●		
Open Binding of Content Identifiers (OBID)			●		●	
X.ig-dw: Implementation guidelines for digital watermarking					●	
DRM technology for digital publicationsPart 1: Overview of copyright protection technologies in …				●		
IEEE Draft Standard for Evaluation Method ofRobustness of Digital Watermarking Implementat…					●	
H.MMAUTH: Framework for authentication of multimedia content		●				
H.274(V4): Versatile supplemental enhancement information messages for coded video bitstreams		●				
H.VADS: Assessment criteria for video authenticity detection services		●				
Credible Web						●
Technical and Governance Guidelines for Responsible Data Collection						●

Standards & Specification Map (cont'd)

	Content Provenance	Trust and Authenticity	Asset Identifiers	Rights Declarations	Watermarking	Other
Data Provenance Standards						●
IEEE Standard for Transparent Human and Machine Agency Identification					●	
IPTC Photo Metadata Standard	●	●		●		
DASH - Part 4: Segment Encryption and Authentication		●				
Recommended Practices for Levels of Artificial Intelligence Generated Content Technologies		●				
CAWG Identity Assertion Specification	●	●				
W3C Verifiable Credentials Data Model v2.0		●				
NIST AI 100-4: Reducing Risks Posed by Synthetic Content		●			●	
MPEG-21 Part 3: Digital Item Identification			●			
Cybersecurity technology—Labeling method for content generated by artificial intelligence	●	●			●	
ITU-T X.2210					●	
HSTP-MMAuth:Multimedia authentication framework, workflows and procedures		●				

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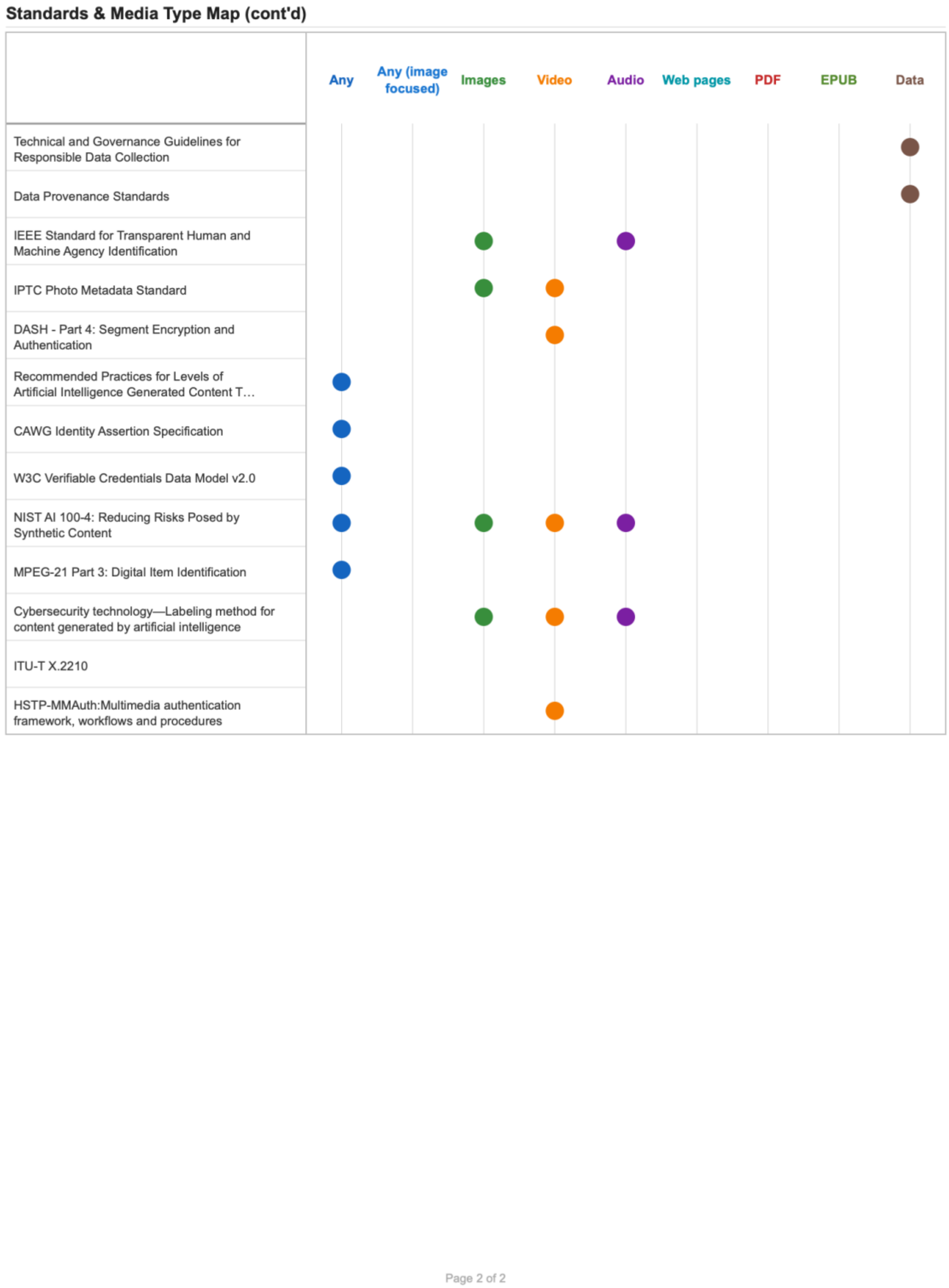
For an accessible table representation of this chart, see Annex A: Table of Standard & Specification Categorization.

Standards & Media Type Map

	Any	Any (image focused)	Images	Video	Audio	Web pages	PDF	EPUB	Data
Content Credentials (C2PA)	●								
JPEG Trust Part 1: Core foundation		●							
JPEG Trust Part 2: Trust profiles catalog		●							
JPEG Trust Part 3: Media asset watermarking			●						
CAWG Metadata	●								
Originator Profile						●			
Overview of trustworthiness in artificial intelligence									
Framework for trust-based media services									
Trust.txt						●			
Chromium Reputation Provider Framework						●			
ISCC: International Standard Content Code (ISCC)	●								
Global Media Identifier (GMI)	●								
TDM Reservation Protocol						●	●	●	
Robots Exclusion Protocol	●								
Vocabulary for expressing AI usage preferences	●								
Open Binding of Content Identifiers (OBID)					●				
X.ig-dw: Implementation guidelines for digital watermarking			●	●					
DRM technology for digital publicationsPart 1: Overview of copyright protection technol…							●	●	
IEEE Draft Standard for Evaluation Method ofRobustness of Digital Watermarking Imple…	●								
H.MMAUTH: Framework for authentication of multimedia content				●					
H.274(V4): Versatile supplemental enhancement information messages for coded …				●					
H.VADS: Assessment criteria for video authenticity detection services				●					
Credible Web						●			



For an accessible table representation of this chart, see Annex B: Table of Media Type Support.

Related Standards

In addition to the core standards discussed in this report, several related areas of standardization play a significant role in supporting digital media authenticity. These standards address foundational technologies and practices that complement the main categories of content provenance, trust and authenticity, asset identifiers, rights declarations, and watermarking.

Blockchain and Distributed Ledger Technologies

Blockchain and Distributed Ledger Technologies (DLT) provide a well-established approach to decentralized, tamper-evident ledgers that can be used to record information about the provenance and integrity of digital assets. By leveraging cryptographic techniques, these technologies enable transparent and immutable records of content creation, modification, and transfer. Standards such as those from ISO TC 307 (Blockchain and distributed ledger technologies) and the W3C (e.g., Verifiable Credentials and DIDs) are increasingly referenced in digital media workflows.

Identity Standards

Robust digital identity standards are essential for establishing the authenticity of content creators, both human and organizational. Efforts emanating from organizations such as the Decentralized Identity Foundation (DIF) and its Creator Assertion Working Group (CAWG, https://cawg.io/identity/1.1/) allow for secure and privacy-preserving identification while enabling features such as attribution of digital assets.

Rights Declarations

It is important to establish a flexible, machine-readable framework for expressing permissions, prohibitions, and obligations associated with digital content. Standards such as the Open Digital Rights Language (ODRL), standardized by the W3C, enable rights holders to specify how their content may be used, shared, or restricted, supporting automated rights management and compliance across diverse platforms and services.

Creative Commons

Creative Commons (CC) licenses are widely adopted for the standardized declaration of usage rights for digital content. The CC framework provides a suite of licenses that allow creators to communicate permissions and restrictions in a clear, interoperable manner.

Supply Chain Integrity, Transparency, and Trust

Supply Chain Integrity, Transparency, and Trust (SCITT) is an emerging initiative led by the IETF, focuses on establishing standards for the secure and transparent tracking of digital assets throughout their lifecycle. SCITT aims to provide mechanisms for recording, verifying, and auditing actions taken on digital content, enhancing supply chain integrity and supporting trust in digital ecosystems. This is particularly relevant for complex media workflows involving multiple stakeholders and distribution channels.

Gap analysis

The standards landscape faces systemic challenges that hinder their efficacy, adoption, and societal alignment. These challenges and gaps can be categorized into governance, access, technical performance, and societal impact. High barriers to entry and non-transparent decision-making processes marginalize diverse stakeholders, while fragmented, paywalled, and IP-encumbered standards complicate implementation for SMEs and researchers. As these standards are adopted, especially at scale, they can struggle with adaptive threats, end-to-end lifecycle security, and consistent privacy protections.

The slow pace of traditional standardization fails to keep up with the rapid evolution of AI technology, leading to fragmented industry solutions. Bridging these gaps requires a move toward agile, open-access, and rights-respecting standards that include sector-specific implementation profiles and rigorous, standardized performance metrics.

Conclusion and next steps

Through the categorization of these existing specifications and standards into key areas, we have highlighted their critical role in fostering trust, accountability, and integrity in the digital ecosystem. The findings underscore the importance of continued collaboration among Standard Development Organizations (SDOs), industry stakeholders, and researchers to address existing gaps and emerging challenges.

As next steps, it is essential to focus on the harmonization of overlapping standards and the development of interoperable frameworks that can be widely adopted across industries. Emerging areas of work, such as the integration of decentralized technologies for enhanced provenance management and the exploration of new watermarking techniques for synthetic media, present exciting opportunities for innovation. Additionally, fostering awareness and adoption of these specifications and standards through education, advocacy, and pilot implementations will be crucial in ensuring their effectiveness and impact.

The evolving nature of digital media and AI technologies necessitates a proactive approach to standardization. By staying ahead of technological advancements and fostering a collaborative ecosystem, we can build a robust foundation for the authenticity and trustworthiness of digital content in the years to come.

Annex A: Table of Standard & Specification Categorization

Specification	Content Provenance	Trust and Authenticity	Asset Identifiers	Rights Declarations	Watermarking	Other
Content Credentials (C2PA)	X	X			X	
JPEG Trust Part 1: Core foundation	X	X	X	X	X	
JPEG Trust Part 2: Trust profiles catalog		X				
JPEG Trust Part 3: Media asset watermarking					X	
CAWG Metadata	X	X		X		
Originator Profile	X	X				
Overview of trustworthiness in artificial intelligence		X				
Framework for trust-based media services		X				
Trust.txt		X		X		
Chromium Reputation Provider Framework		X				
ISCC: International Standard Content Code (ISCC)			X			
Global Media Identifier (GMI)			X			
TDM Reservation Protocol				X		
Robots Exclusion Protocol				X		
Vocabulary for expressing AI usage preferences				X		
Open Binding of Content Identifiers (OBID)			X		X	
X.ig-dw: Implementation guidelines for digital watermarking					X	
DRM technology for digital publications						
Part 1: Overview of copyright				X		

protection technologies in use in the publishing industry						
IEEE Draft Standard for Evaluation Method of						
Robustness of Digital Watermarking Implementation in Digital Contents					X	
H.MMAUTH:						
Framework for authentication of multimedia content		X				
H.274(V4):						
Versatile supplemental enhancement information messages for coded video bitstreams		X				
H.VADS:						
Assessment criteria for video authenticity detection services		X				
Credible Web						X
Technical and Governance Guidelines for Responsible Data Collection						X
Data Provenance Standards						X
IEEE Standard for Transparent Human and Machine Agency Identification					X	
IPTC Photo Metadata Standard	X	X		X		
DASH - Part 4: Segment Encryption and Authentication		X				
Recommended Practices for Levels of Artificial Intelligence Generated Content Technologies		X				

CAWG Identity Assertion Specification	X	X				
W3C Verifiable Credentials Data Model v2.0		X				
NIST AI 100-4: Reducing Risks Posed by Synthetic Content		X			X	
MPEG-21 Part 3: Digital Item Identification			X			
Cybersecurity technology—Labeling method for content generated by artificial intelligence	X	X			X	
ITU-T X.2210					X	
HSTP-MMAuth: Multimedia authentication framework, workflows and procedures		X				

Annex B: Table of Media Type Support

Standard	Audio	Data	EPUB	Images	PDF	Video	Web pages	Others
Content Credentials (C2PA)	X	X	X	X	X	X	X	X
JPEG Trust Part 1: Core foundation	X	X	X	X	X	X	X	X
JPEG Trust Part 2: Trust profiles catalog	X	X	X	X	X	X	X	X
JPEG Trust Part 3: Media asset watermarking				X				
CAWG Metadata	X	X	X	X	X	X	X	X
Originator Profile							X	
Overview of trustworthiness in artificial intelligence								
Framework for trust-based media services								
Trust.txt							X	
Chromium Reputation Provider Framework							X	
ISCC: International Standard Content Code (ISCC)	X	X	X	X	X	X	X	X
Global Media Identifier (GMI)	X	X	X	X	X	X	X	X
TDM Reservation Protocol			X		X		X	
Robots Exclusion Protocol	X	X	X	X	X	X	X	X
Vocabulary for expressing AI	X	X	X	X	X	X	X	X

usage preferences								
Open Binding of Content Identifiers (OBID)	X							
X.ig-dw: Implementation guidelines for digital watermarking				X		X		
DRM technology for digital publications								
Part 1: Overview of copyright protection technologies in use in the publishing industry			X		X			
IEEE Draft Standard for Evaluation Method of								
Robustness of Digital Watermarking Implementation in Digital Contents	X	X	X	X	X	X	X	X
H.MMAUTH:								
Framework for authentication of multimedia content						X		
H.274(V4):								
Versatile supplemental enhancement information messages for coded video bitstreams						X		
H.VADS:								

Assessment criteria for video authenticity detection services						X		
Credible Web							X	
Technical and Governance Guidelines for Responsible Data Collection		X						
Data Provenance Standards		X						
IEEE Standard for Transparent Human and Machine Agency Identification	X			X				
IPTC Photo Metadata Standard				X		X		
DASH - Part 4: Segment Encryption and Authentication						X		
Recommended Practices for Levels of Artificial Intelligence Generated Content Technologies	X	X	X	X	X	X	X	X
CAWG Identity Assertion Specification	X	X	X	X	X	X	X	X
W3C Verifiable Credentials Data Model v2.0	X	X	X	X	X	X	X	X
NIST AI 100-4: Reducing Risks Posed by Synthetic Content	X	X	X	X	X	X	X	X

MPEG-21 Part 3: Digital Item Identification	X	X	X	X	X	X	X	X
Cybersecurity technology—Labeling method for content generated by artificial intelligence	X			X		X		
ITU-T X.2210						X		
HSTP-MMAuth:						X		

Annex C: Table of Summary of Identified Gaps

Gap Category	Primary Issues
Participation & Governance	High barriers for SMEs/civil society, lack of transparency in decision-making, and insufficient inclusion of non-technical human rights perspectives.
Access & Usability	Paywalled documentation, standard fragmentation, IP/licensing burdens, and lack of sector-specific implementation profiles.
Measurement & Assurance	Lack of standardized performance metrics, insufficient conformance testing, and scarcity of open reference implementations.
Technical Gaps	Weak end-to-end security, inadequate threat modeling against adaptive adversaries, and inconsistent privacy/data minimization.
Societal & Operational	Poor integration of human rights/legal requirements and under-standardized UI/UX patterns.
Speed & Agility	Slow standardization processes compared to technology evolution and lack of experimental "sandbox" standards.